Morteza Abedi(He/Him)

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Research Interests

Federated Learning, Medical Artificial Intelligence, Deep Learning, Adversarial Machine Learning, Biomedical Signal Processing, Healthcare Analytics, Transfer Learning, Computer Vision, Rare Disease Diagnosis.

Education

M.Sc. Computer Engineering (Artificial Intelligence and Robotics)

Institute for Advanced Studies in Basic Sciences (IASBS), Iran (2021 – 2024)

Thesis: *Strengthening the Resistance of Electrocardiographic Signals Against Aggressive Attacks Using Conditional Adversarial Networks (CGAN)*, GPA: 17.43 / 20

Supervisor: Prof. Mahdi Bohlouli

B.Sc. Computer Engineering (Information Technology)

University of Gonabad, Iran (2015 – 2019)

Thesis: *Recognizing Handwritten Numbers and Emotional States of Human Faces Using Deep Learning*

Supervisor: Dr. Havva Alizadeh Noughabi

Publications

Journal Article

Abedi, M., & Bohlouli, M.

TL-CGAN: How Transfer Learning Conditional GAN Makes Robustness of Electrocardiogram (ECG) Signals Against Adversarial Attacks

PLOS ONE – Under Revision (2025)

Research Experience

Administrative and Public Relations Expert (Military Obligatory Service – Amriye)
Industry and Society Relations Office, IASBS Feb 2025 – Jun 2026

- Coordinated between different university units for planning and execution of official events and programs.
- Collected, organized, and updated organizational information for publication on the university website and newsletter.
- Handled official correspondence and followed up administrative matters with internal units and external organizations.

Senior Research Assistant

Intelligent Systems Group (ISG), IASBS — Jan 2021 – Feb 2024

- Conducted research in Artificial Intelligence and Machine Learning for healthcare applications.
- Developed Deep Learning solutions for biomedical signal processing.
- Investigated adversarial robustness of ECG-based systems.
- Collaborated in interdisciplinary AI projects involving healthcare data.

Research Staff

IoTrain Erasmus+ Project — Sep 2022 – Dec 2024

- Led Work Package 4 as part of the IASBS team, in collaboration with international partners.
- Contributed to international IoT-focused research activities and project deliverables.
- Participated in quality monitoring processes and collaborated with academic partners across Europe.

Teaching Experience

Lecturer Assistant, IASBS

Machine Learning and Deep Learning in IoT (Apr 2024 – Sep 2024)

Teaching Assistant, IASBS

- Distributed Systems (Winter 2024)
- Intelligent Systems and Applications in IoT (Fall 2023)

Academic Service

Reviewer

- PLOS ONE, Biomedical Signal Processing and Control, Expert Systems with Applications, ACM International Conference on Information and Knowledge Management(CIKM), The 18th International Symposium on Visual Information Communication and Interaction (VINCI)

Selected Projects

- **Anomaly Detection in Organizational Requests** (Oracle APEX) – Detected anomalies based on user request volume, calculated anomaly scores, and identified behavioral patterns in the database (2026).
- **Behavioral Anomaly Detection in Autism Patient Data** (2025).
- **Anomaly Detection of Critical Components in Industrial Production Line Data** (2025).
- **Federated Learning Models for Optimization and Security of Sensitive Industrial Data** (2025).
- **Medical Signal Processing for Detection of Specific Disease Anomalies** (2023–2024).
- **Time-Series Data Preparation and Analysis for AI Algorithms** (2022).
- **Implementation of Machine Learning and Deep Learning algorithms on standard datasets (MNIST, CIFAR) for classification, clustering, and image processing** (2021).
- **Twitter data collection and analysis pipeline using Natural Language Processing (NLP)** (2020).

Honors and Awards

- Winner, Research and Technology Achievements Provincial Exhibition Award (2022)
- Scholarship Recipient, International Winter School on IoT (2023)
- Visiting Scholar, Staff Training and Mobility Event, University of Siegen, Germany (2022)
- Technical Staff Member, International Summer School on IoT, Erasmus+ IoTrain Project (2024)
- Member, Industrial Potentials of IoT in Main Industries, IoTrain Erasmus+ Project (2023)
- Member, Quality Monitoring Committee, IoTrain Erasmus+ Project, Bucharest, Romania (2023)

Technical and Research Skills

Research Expertise

Machine Learning, Deep Learning, Federated Learning, Medical AI, Biomedical Signal Processing, Adversarial Machine Learning, Transfer Learning, Computer Vision

Programming and Frameworks

Python, PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, YOLO, FedAvg, MATLAB, C++

Research Tools

LaTeX, Git, Docker, SQL, MySQL, NoSQL, Oracle APEX

Languages

Persian (Native)

English (Upper-Intermediate, Professional Working Proficiency)

References

Available upon request